

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Lecture 2: Measures of Model Performance

Measure of Model Performance: Topics

- Linear algebra review
- Supervised model introduction
- Metrics of performance
- Be able to create receiver operating characteristic curves
- **kNN – a simple supervised learning model**

K-Nearest Neighbor: Example

Recognizing types of Iris flowers (by R. Fisher)



setosa



versicolor

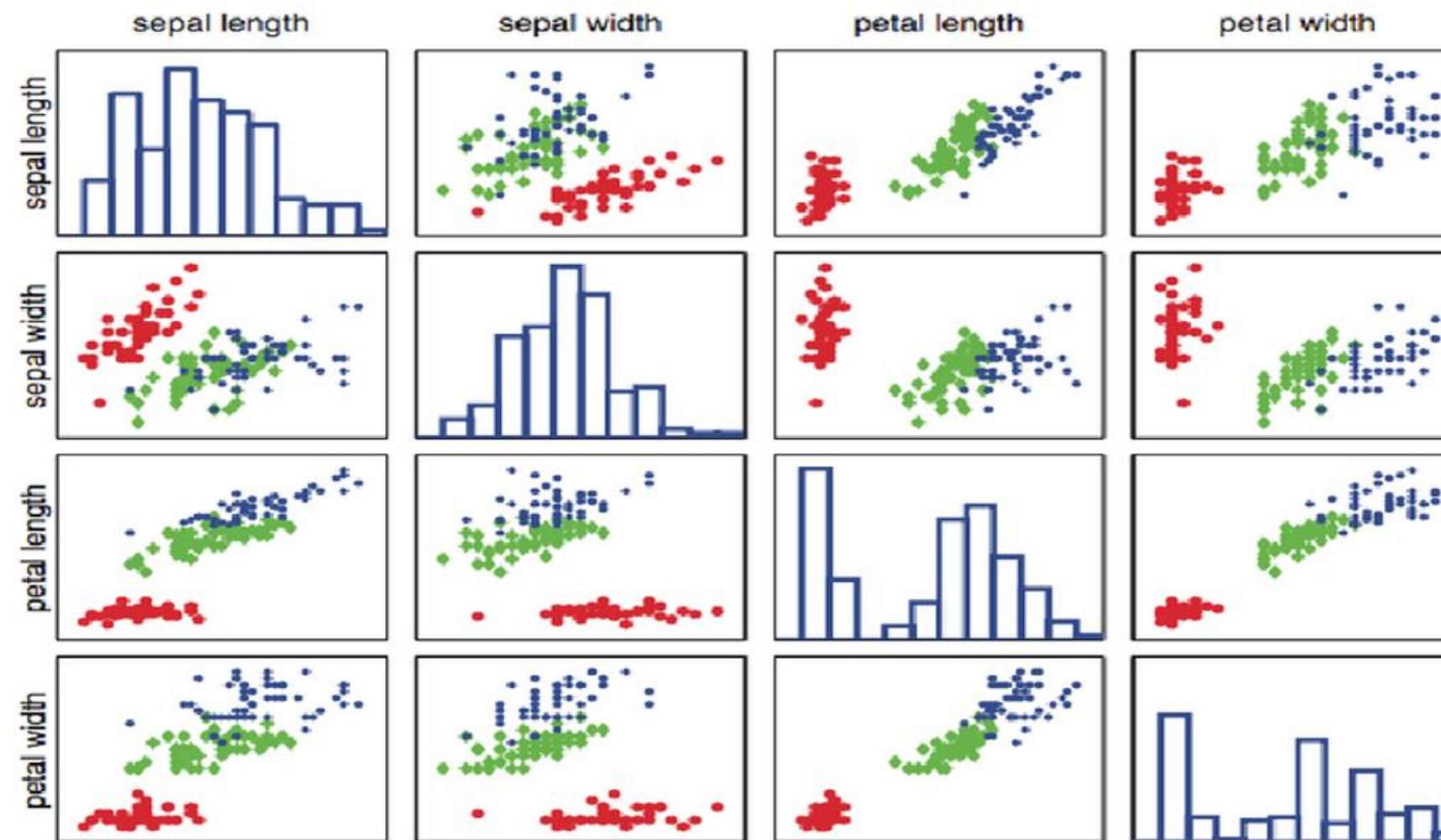


virginica

Features: the widths and lengths of sepal and petal

K-Nearest Neighbor: Examples (1)

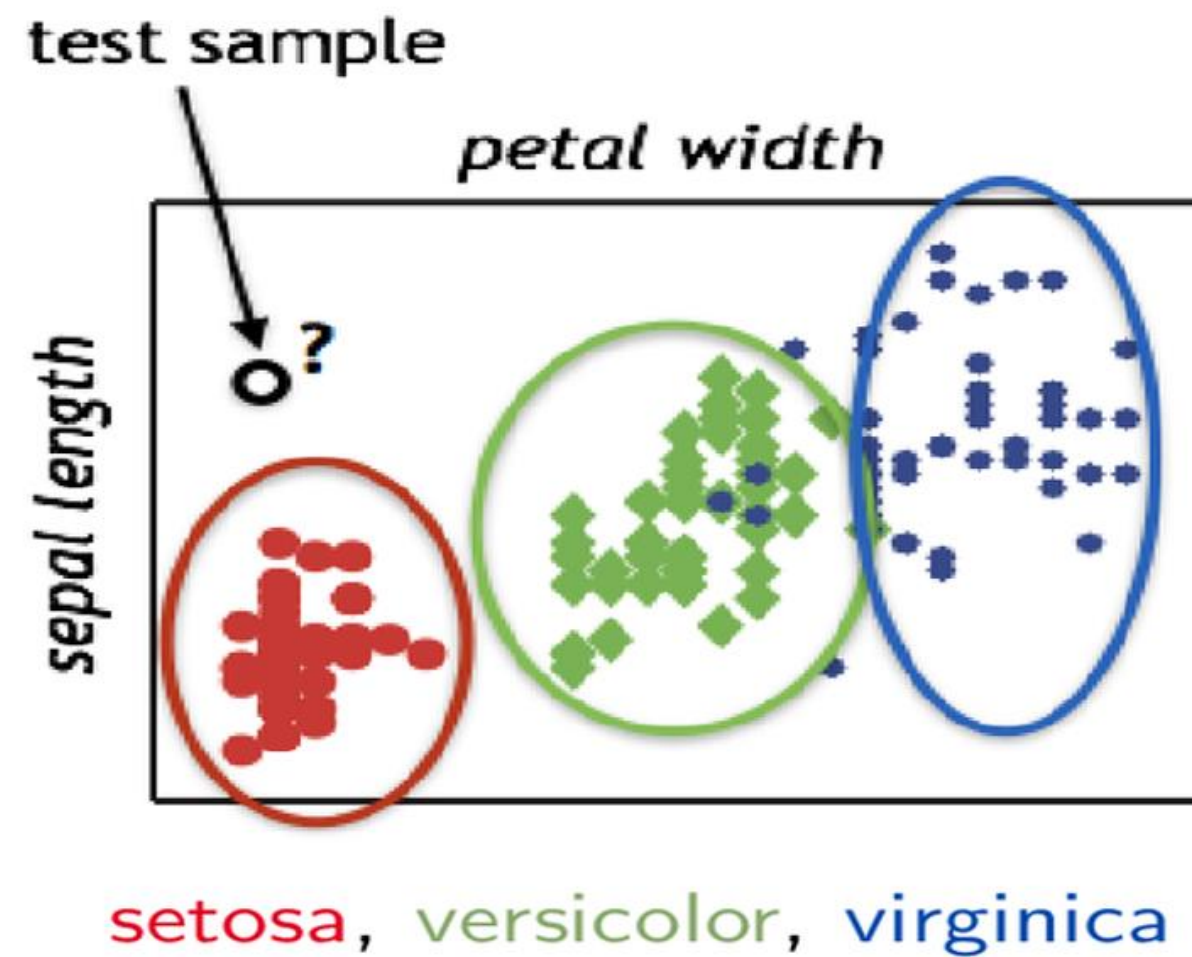
Visualizing features to get better intuition about our data



Each colored datapoint is one sample
setosa, versicolor, virginica

K-Nearest Neighbor: Examples (2)

Using two features: sepal length & petal width



Test sample is closer to red cluster → label it as **setosa**

K-Nearest Neighbor: Examples (3)

Recognizing types of Iris flowers (by R. Fisher)

Often data is organized in a table

Each row is one sample with 4 features and 1 label

```
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
4.7,3.2,1.3,0.2,Iris-setosa
4.6,3.1,1.5,0.2,Iris-setosa
5.0,3.6,1.4,0.2,Iris-setosa
5.4,3.9,1.7,0.4,Iris-setosa
4.6,3.4,1.4,0.3,Iris-setosa
5.0,3.4,1.5,0.2,Iris-setosa
4.4,2.9,1.4,0.2,Iris-setosa
4.9,3.1,1.5,0.1,Iris-setosa
5.4,3.7,1.5,0.2,Iris-setosa
4.8,3.4,1.6,0.2,Iris-setosa
4.8,3.0,1.4,0.1,Iris-setosa
4.3,3.0,1.1,0.1,Iris-setosa
```

Attribute Information:

1. sepal length in cm
2. sepal width in cm
3. petal length in cm
4. petal width in cm
5. class:
 - Iris Setosa
 - Iris Versicolour
 - Iris Virginica

[Source: <https://archive.ics.uci.edu/ml/datasets/iris>]

K-Nearest Neighbor: Representation (4)

- Training Data

- N samples/datapoints/instances: $S^{train} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$
- Used for learning representation: $f: \mathbf{x} \rightarrow y$

- Testing Data

- M samples/datapoints/instances: $S^{test} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_M, y_M)\}$
- Used to assess how well $f(\cdot)$ will do in predicting an unseen sample

- Train and test data should not overlap: $S^{train} \cap S^{test} = \emptyset$

K-Nearest Neighbor: Representation (5)

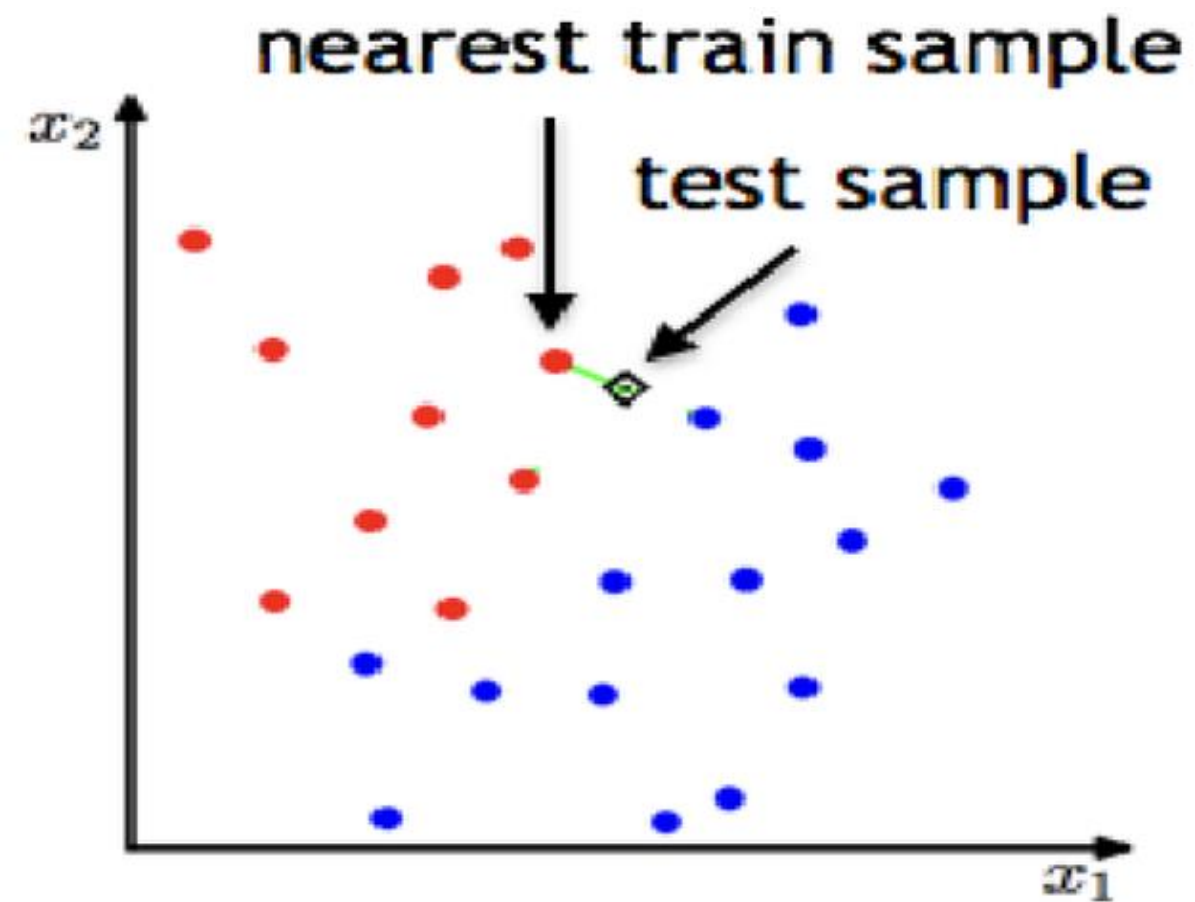
- Classify data into one out of multiple classes
 - Input: $\mathbf{x} \in \mathbb{R}^D$
 - Output: $y \in \{1, 2, \dots, C\}$, C is the number of classes
 - Model: $f: \mathbf{x} \rightarrow y$
- Special case: binary classification ($C=2$)
 - Output: $y \in \{1, 2\}$ or $\{0, 1\}$ or $\{-1, 1\}$

K-Nearest Neighbor: Representation (6)

- Nearest Neighbor (or 1-Nearest Neighbor, 1-NN)
 - Assigns test sample $\mathbf{x}'$ to the closest training sample
 - Model:
 - $\hat{y} = f(\mathbf{x}) = \mathbf{y}_{nn(\mathbf{x})}$
 - $nn(\mathbf{x}) = \arg \min_{i=1, \dots, M} \|\mathbf{x}' - \mathbf{x}_i\|^2 = \arg \min_{i=1, \dots, M} \sum_{j=1}^D (\mathbf{x}'_j - \mathbf{x}_{ij})^2$

K-Nearest Neighbor: Representation (7)

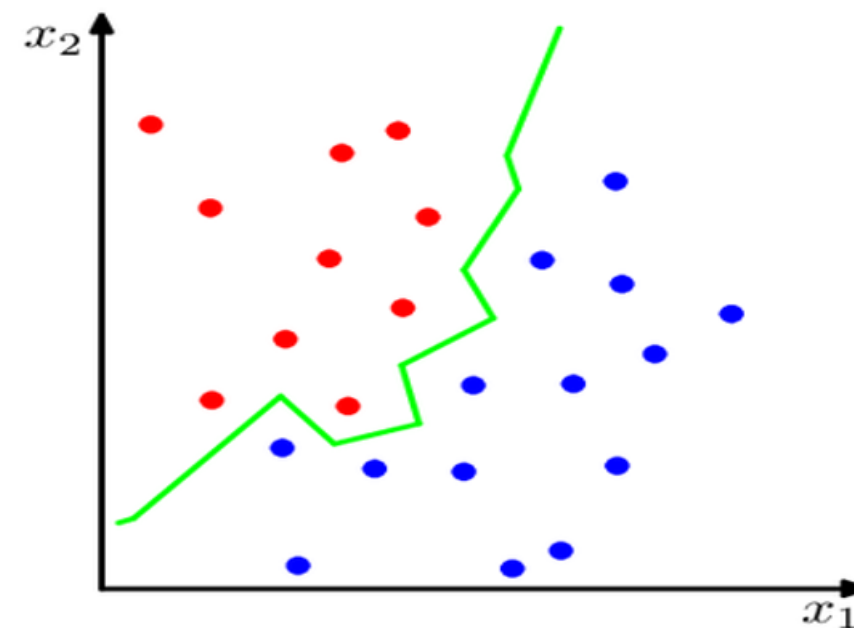
- Nearest Neighbor (or 1-Nearest Neighbor, 1-NN): Example



- The nearest point to test sample x' is a red training instance, therefore x' will be labeled as red

K-Nearest Neighbor: Representation (8)

- Nearest Neighbor (or 1-Nearest Neighbor, 1-NN): Example
 - Decision boundary: For every point in the space, we can determine its label using the nearest neighbor rule. This gives us a decision boundary that partitions the space into different regions.



- The above decision boundary is very sensitive to noise. What would be the solution for this?

K-Nearest Neighbor: Representation (9)

Increase number of nearest neighbors to use

- 1-nearest neighbor: $nn_1(\mathbf{x}') = \arg \min_{i \in \{1, \dots, M\}} \|\mathbf{x}' - \mathbf{x}_i\|_2^2$
- 2-nearest neighbor: $nn_2(\mathbf{x}') = \arg \min_{i \in \{1, \dots, M\} \setminus nn_1(\mathbf{x}')} \|\mathbf{x}' - \mathbf{x}_i\|_2^2$
- 3-nearest neighbor: $nn_3(\mathbf{x}') = \arg \min_{i \in \{1, \dots, M\} \setminus \{nn_1(\mathbf{x}'), nn_2(\mathbf{x}')\}} \|\mathbf{x}' - \mathbf{x}_i\|_2^2$

The set of K-nearest neighbors is

$$knn(\mathbf{x}) = \{nn_1(\mathbf{x}'), \dots, nn_K(\mathbf{x}')\}$$

Neighbors $nn_1, \dots, nn_K$ in order of increasing distance from sample $\mathbf{x}'$

K-Nearest Neighbor: Representation (10)

- K-NN Model

- Each neighbor in $knn(\mathbf{x}) = \{nn_1(\mathbf{x}), \dots, nn_K(\mathbf{x})\}$ votes one class
- Count the number of neighbors that have voted each class (C)

$$v_c = \sum_{k \in knn(\mathbf{x})} I_{[y_k=c]}, c = 1, \dots, C$$

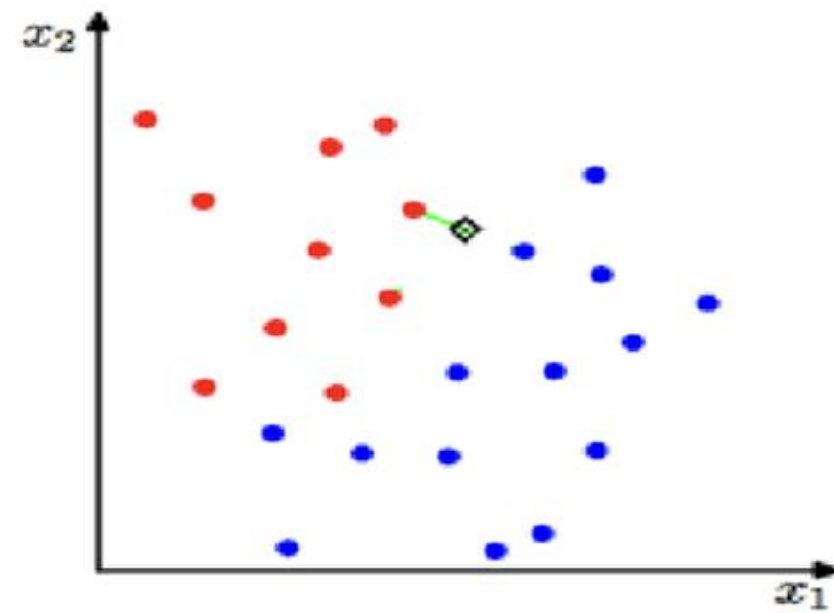
- Assign test sample $\mathbf{x}$ to the majority class membership of the K neighbors

$$y = f(\mathbf{x}) = \arg \max_{c=1, \dots, C} v_c$$

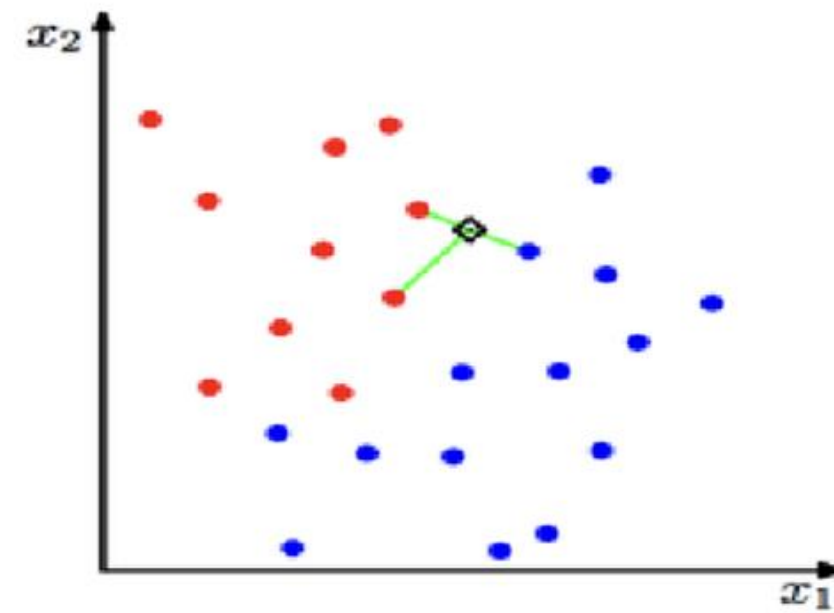
K-Nearest Neighbor: Representation (11)

- K-NN Example

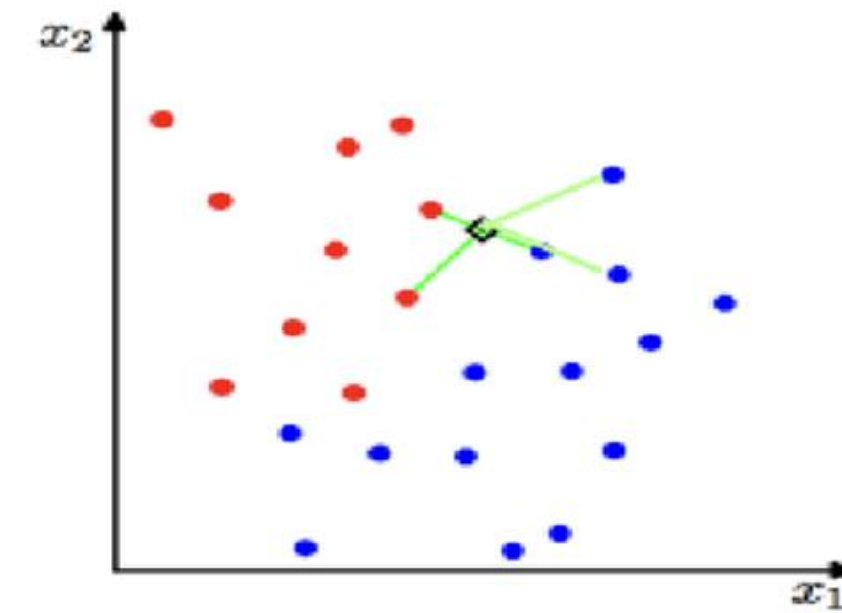
K=1, label=red



K=3 label=red

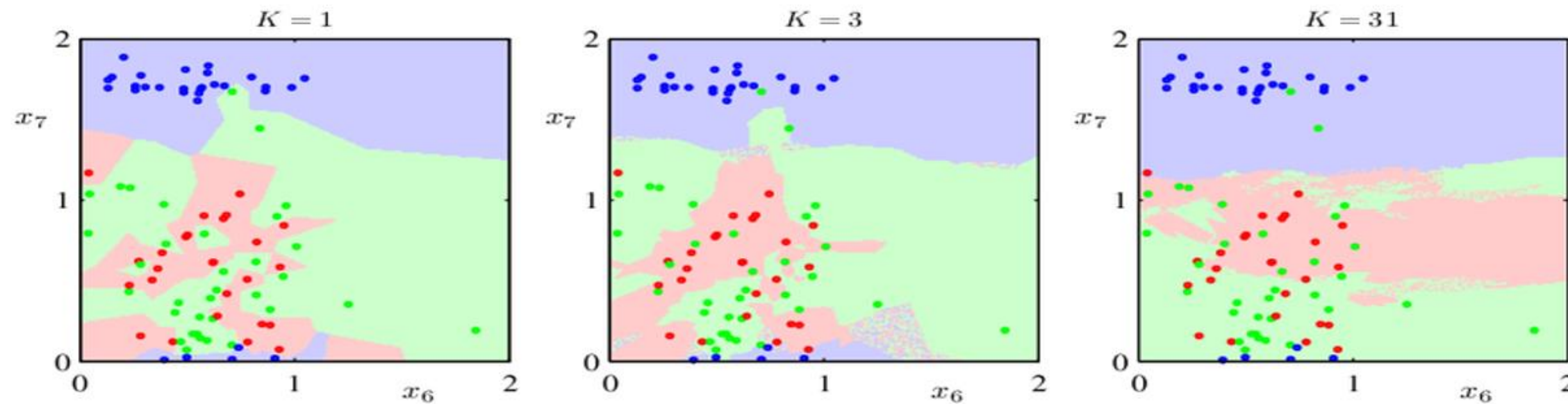


K=5, label=blue



K-Nearest Neighbor: Representation (12)

K-NN Decision Boundary



Number of neighbors K controls the degree of smoothing

- $K \downarrow$: many small regions of each class
- $K \uparrow$: fewer larger regions of each class

Measure of Model Performance: Topics (End)

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